

IoT Service Orchestration in Edge–Cloud Continuum With 6G: A Review

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Abstract—The development of sixth-generation (6G) mobile networks brings advancements in wireless communications, including lower latency, higher data rates, and improved spectral efficiency, as well as enhanced facilitation for the use of artificial intelligence (AI) technologies, integrated communications, sensing, and renewable energy sources. To harness this potential, a scalable framework integrating edge computing (EC) and cloud computing (CC) is needed to provide a robust computing continuum for various Internet of Things (IoT) applications and services. This article analyses the strengths and weaknesses of existing and emerging distributed computing frameworks—ranging from traditional centralized IoT architectures, through fog, edge, and local-edge architectures, to the full three-tier edge–cloud continuum—in terms of the IoT service orchestration and diverse application requirements. We emphasize the latest evolutionary step, the three-tier edge–cloud continuum, which aims to resolve the most significant limitations of the previous frameworks, recognized in the literature. It enables efficient IoT service orchestration by utilizing distributed resources and computation closer to end users, while taking full advantage of the centralized resources at data centers. We evaluate the maturity of these frameworks, considering factors such as scalability, resource efficiency, adaptability to changes, resource availability, security, and privacy, as well as robustness and resilience. Overall, this study aims to serve as a roadmap for researchers, network architects, and industry stakeholders to make informed decisions on implementing the computing continuum in 6G networks.

Index Terms—Computing continuum, edge–cloud continuum, far edge, fog computing (FC), Internet of Things (IoT) applications, local edge, multiaccess edge computing (EC), sixth-generation (6G).

NOMENCLATURE

AI	Artificial intelligence.
AR	Augmented reality.

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CC	Cloud computing.
C-RAN	Cloud-radio access network.
CPU	Central processing unit.
DL	Deep learning.
EC	Edge computing.
FC	Fog computing.
GPU	Graphics processing unit.
IoT	Internet of Things.
KPIs	Key performance indicators.
MEC	Multiaccess edge computing.
ML	Machine learning.
NFV	Network function virtualization.
PaaS	Platform as a service.
QoS	Quality of service.
RAN	Radio access network.
VR	Virtual reality.
ISAC	Integrated sensing and communication.
JCAS	Joint communication and sensing.

I. INTRODUCTION

OVER the past decade, most organizations have moved their enterprise applications to the cloud for better performance, availability, scalability, elasticity, and maintainability of resources [1], [2], [3]. CC has traditionally been emphasized in the IoT applications, where constrained-capacity local nodes do not provide sufficient resources for various computational needs. Despite the indisputable benefits of CC, the traditional data center-based architecture faces challenges, such as high latency, sensitivity to network problems, and inefficient use of network resources. These drawbacks have been identified as major bottlenecks for both data-intensive and real-time IoT applications and services [4]. To address this, the concepts of EC and FC, bringing cloud-integrated computational resources and processing closer to the data sources and end users, have been proposed. The shorter distance between data sources and users with computing resources helps mitigate the challenges of CC by reducing network latency, providing more efficient use of network resources, and improving both failure tolerance and security.

Various definitions for these technologies and their relationship have been made. EC is important for applications that require lower latency, instant and real-time response, and high bandwidth. However, unlike CC, where data computation is centralized on remote servers, EC runs data processing closer to the source of data, for instance, at base stations or

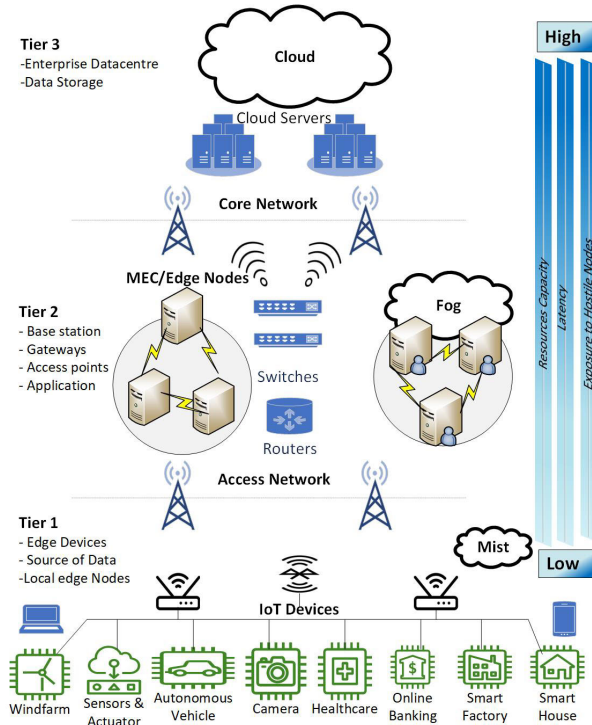


Fig. 1. Three-tier edge-cloud continuum: tier 1—IoT devices and local EC; tier 2—EC and FC nodes; and tier 3—cloud data centers.

the IoT gateways. The proposed architecture of OpenFog [5], [6], for example, considers EC as a pool of computational resources located on intermediate nodes situated between IoT sensors/actuators and the cloud infrastructure. Whereas EC is seen as referring to the computational tier between the local nodes and data centers, FC is described as the processing continuum facilitating the end-to-end execution of tasks spanning from the IoT devices to cloud data centers [3], [6], [7]. Thus, FC can be seen as a continuum of processing and computation of data from the cloud to the edge and to the IoT [8].

A distributed computing architecture, which combines the devices at the local edge tier and access edge tier, including EC and FC, and data centers at the cloud tier, including CC, is known as the three-tier edge-cloud continuum, as illustrated in Fig. 1. Depending on the proximity to the source of data, each computing tier offers a distinct set of processing power and storage, permitting effective data processing. This article explores the various existing architectural frameworks for this computing continuum, particularly examining three computing tiers that can be provisioned either independently or integrated into a unified logical platform [9]. In this article, we aim to examine the origins, motivations, and level of development of alternative architectural frameworks for the computing continuum. This article will additionally provide a comparative analysis of orchestration mechanisms for managing the dynamic resource allocation and deploying various tasks in the sixth-generation (6G) computing continuum. The three-tier edge-cloud continuum architecture discussed in this article is presented in Fig. 1 and explained in the following.

Tier 1 (the bottom tier in Fig. 1) consists of local computing nodes with limited resource capacities, edge devices, and swarms of connected IoT devices. To ensure instant response and quick processing, tier 1 typically manages the time-critical and latency-sensitive IoT services. Due to its lightweight architecture, this tier provides EC and FC capacity for lightweight computing operations, requiring high availability and low latency. It enables effective orchestration for critical task execution and runtime decision-making with limited hardware resources.

Tier 2 (the mid-tier in Fig. 1) consists of integrated base stations, access points, gateways, and dedicated EC and FC nodes. Tier 2 resourcefully deploys resource-intensive and latency-sensitive IoT services and applications to the computing nodes. Tier 2 offers distributed data processing and computation with a greater resource capacity than tier 1. Tier 2 undertakes advanced scalability and reliability to accommodate massive IoT service orchestration. The proximity of multiaccess edge computing (MEC) and other EC and FC nodes with the data source helps reduce the latency and balance workload distribution to the computing nodes.

Tier 3 (the top tier in Fig. 1) includes globally available enterprise data centers and cloud servers with vast resources and data storage capacity. This tier is optimal for resource-intensive, nonlatency-sensitive IoT services and applications, including large-scale computing and data storage. Tier 3 enables the global orchestration of IoT services and coordinated resource allocation between all tiers.

The integration of the multiple computing tiers indicated above is called as the edge-cloud continuum. Based on requirements related to latency, resource availability, and an increasing number of application requests, it ensures that data and computational tasks can, whenever needed, migrate quickly across the various tiers. The continuum dynamically distributes workloads and data throughout the dispersed infrastructure to maximize the network's overall performance, responsiveness, and resource consumption. Fig. 1 represents the hierarchy of the tiers. Tier 1, the lowermost tier, encompasses the established massive IoT devices and serves as the primary data source. Tier 2 consists of EC, MEC, and FC nodes. Infrastructure components like base stations, gateways, and access points are crucial in managing and orchestrating numerous services and applications. Tier 3 illustrates the core network architecture with cloud servers or data centers hosting CC. Fig. 2 represents the placement of five architectural frameworks identified from the literature. Related work is presented in Table I.

The advancement of network technologies from the second generation (2G) toward the 6G networks implies continuous success in network connectivity, service advancement, and intelligence of the network. Prior generations, such as 2G and 3G were mainly positioned on digital voice calling and basic data services and supported text messaging. Later, the fourth-generation (4G) networks pioneered the mobile broadband networks and offered relatively high data throughput that backed massive multimedia and mobile applications. The fifth-generation (5G) systems enhanced the network by bringing significantly higher data rates, ultrareliable and low-latency

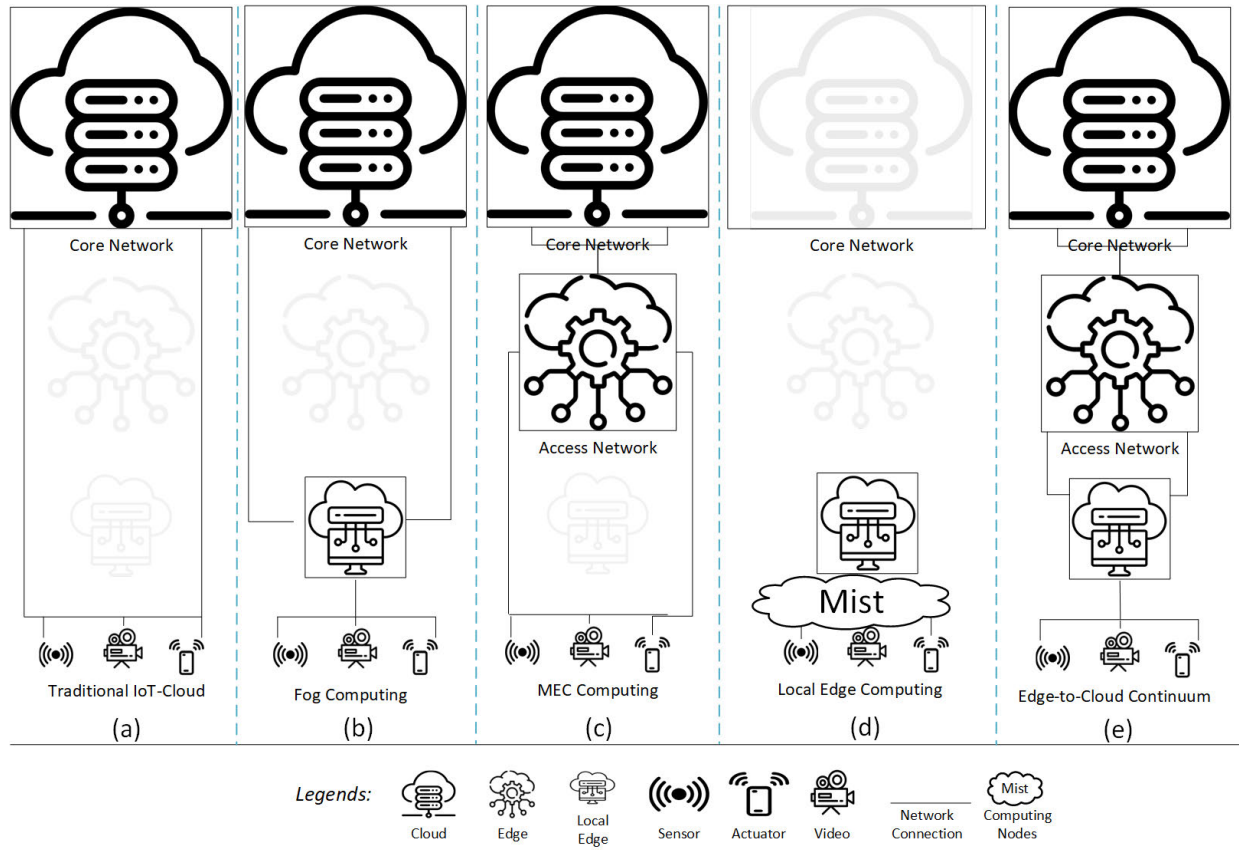


Fig. 2. IoT service orchestration to computing nodes: (a) centralized IoT-CC; (b) FC; (c) multiaccess EC; (d) local EC; and (e) three-tier edge-cloud continuum.

communication (URLLC), massive machine-type communications (mMTC), and service deployment by applying network slicing and MEC for the acceptance of IoT applications and services [10].

The 6G networks aim to extend these performance metrics by developing terahertz communications, extensive AI integration, and hyperreliable and low-latency communication (HRLLC) connectivity for the advancement of Internet of Everything (IoE) ecosystems [11]. The IoE ecosystem extends the conventional IoT by integrating individuals, user data, processes, and intelligent things into an interconnected and data-centric cyber-physical-social system, enabling real-time awareness, decision-making, reasoning, and autonomous systems. IoE is adopting AI-driven networking, HRLLC, and edge AI to provide massive, context-driven, and time-critical applications in smart cities, healthcare, and industrial automation in 6G and future network technologies.

These developments aim to enhance high data throughput, offer HRLLC communication, network connection density, and bring intelligence to the edge, which enables extraordinary support for autonomous systems, semantic communications, and intelligent service orchestration across the edge-cloud continuum through the convergence of communications and sensing ISAC [12] and JCAS [13] and the convergence of communications and computing with integrated AI/ML [14].

This article is organized as follows. Section II provides a brief overview of the edge-cloud continuum frameworks

proposed in the literature, supported by Fig. 2, and concentrates on related work, investigating the shortcomings and gaps in the existing literature. Section III signifies the core of this article, where the five key topics of this article are discussed in depth. The computing continuum is analyzed from the local edge to cloud tiers, emphasizing the orchestration of IoT applications and services. Section IV provides an overview of edge-enabled AI applications in 6G, explaining the placement of AI-driven applications within the edge-cloud continuum and emphasizing AI-specific attributes. Section V summarizes the findings, highlighting key insights and offering future research directions for the academic community. Finally, Section VI presents the conclusion, combining the significant takeaways and highlighting this article's contributions to the field.

II. BACKGROUND AND RELATED WORK

In this section, we identify and compare five existing frameworks from the literature for managing the computing continuum and analyze their features, benefits, and drawbacks from the 6G systems and applications perspective. First, we investigate the communication and computing processes within the traditional IoT-cloud architecture. Next, we discuss the concepts of FC and MEC within the continuum. Lastly, the significance of implementing computations at the correct tier in the edge-cloud continuum is described in detail.

TABLE I
RELATED WORK

Year	Ref.	Short Summary	Orc	DC	RM	NS	NSec	Sc
2025	[41]	The survey paper investigates resource scheduling in MEC leveraging RL and DRL ML techniques. It explores mobility-related problems and discusses open issues & challenges.	✓	✓	✓	✓	✓	✓
2024	[40]	A review of MEC is provided, including its background and how it differs from other computing paradigms. In addition to highlighting current advancements in computation offloading research, it also explores the architecture of MEC systems through an example of a typical use case.	-	✓	✓	-	✓	-
2024	[30]	Address the impact of AI and collective research on bringing communication, computation, and caching (3C) resources and services in the cloud-to-edge continuum.	✓	✓	-	✓	-	-
2024	[32]	This article proposed a system model for Edge-Cloud continuum integration and classifies federated learning applications in cloud and edge perspectives. Highlight the important research gaps and implementation issues of federated learning for IoV.	-	✓	✓	-	-	✓
2024	[33]	Analyze the challenges of edge-enabled IoT applications in adaptive EC systems.	-	✓	-	-	-	✓
2024	[34]	This survey delves into the evolution from 5G to 6G networks, emphasizing the critical aspects of security and privacy. The authors identify potential risks associated with the integration of emerging technologies such as IoT, EC, and AI-driven analytics within 6G infrastructures.	-	-	✓	-	✓	✓
2023	[31]	This survey explores the current state and challenges of IoT-Edge-Cloud continuum systems, focusing on the orchestration and security aspects across different tiers. The authors discuss the necessity of distributing security measures throughout the continuum to address the varying computational capacities and threat landscapes inherent in each tier.	✓	-	✓	-	✓	-
2023	[36]	Investigate resource management methods for service orchestration in EC architecture with standardization and benefits.	✓	✓	✓	✓	-	-
2023	[31]	Addressing the extensive challenges like connection protocols for diverse devices, task offloading, security and privacy improvements against threats, scalability and mobility, and ideal resource allocation to resources in IoT-Edge-Cloud (IEC) continuum systems.	-	✓	-	-	-	✓
2022	[20]	A comprehensive review of MEC development, its challenges to reach applications and service requirements like ULL and BW, and the emerging technologies supporting MEC such as ICN, NFV, SDN, SFC, C-RAN, and F-RAN.	✓	✓	-	✓	-	-
2022	[37]	Presents literature that focuses on designing and implementing the MEC infrastructure using NFV to enhance service development and resource management for both virtual and physical networks.	✓	✓	✓	-	✓	✓
2021	[38]	Describes the EC architecture using a unified model to emphasize the challenges of resource optimization through resource scheduling and provisioning techniques.	✓	✓	✓	-	✓	✓
2019	[3]	Highlighting the effect of mobile CC on Cloudlet, EC, and mobile Ad hoc cloud development, this article also presents the state-of-the-art and future challenges of decentralized cloud models.	✓	✓	✓	-	-	✓
2018	[39]	Outlines the taxonomy of concepts within FC and EC architecture. Defines key parameters to highlight the differences among existing fog systems. Discusses the logical and physical ecosystems of the cloud, fog, and edge architecture.	✓	✓	✓	-	✓	✓
This work	-	Service orchestration in three-tier computing continuum i.e. traditional Iot-cloud, FC, MEC, EC, and local edge	✓	✓	✓	✓	✓	✓

A. Existing and Emerging IoT Computing Frameworks

1) *Centralized IoT-CC*: Traditional cloud-based IoT [Fig. 2(a)] serves as the baseline architecture for this study, consisting of local sensors, actuators, user interface (UI) nodes, and remote cloud-based services. In this architecture, services are deployed on one or more data centers, depending on the individual service structure, while the related sensors and actuators are located in the physical locations where the services are used. The basic limitation of this approach is the connectivity between the local devices and the cloud, set over the Internet. This structure is not inherently suitable for applications requiring high reliability, exchanging large amounts of data between local nodes and the server, or requiring low latency [15]. However, the traditional centralized architecture does not comprehensively support critical low-latency IoT systems and applications. Consequently, the use of edge and FC has been proposed as a potential solution to address the constraints associated with CC [16], [17].

2) *Fog Computing*: The FC paradigm [Fig. 2(b)] involves the execution of computing tasks and data processing at the

edge of the network, typically at the IoT gateways or other dedicated local computing nodes. The goal for FC is to extend cloud features and functionalities such as storage, databases, and computing along with services like building automation to the edge of the network, closer to users and data sources. In FC, a computational task requiring, e.g., low response time and high reliability, could be run on the edge, while tasks that are delay-tolerant and computationally intensive would deploy to servers in data centers [16], [17]. The gained benefits include latency, resilience for network errors, and improved security [18], [19]. The following are the features offered by the FC paradigm:

- 1) provisioning/deprovisioning resources based on the need;
- 2) provisioning storage of data at a small scale;
- 3) flexibility of computation deployment (cloud and local);
- 4) scalability (support for large-scale deployments);
- 5) facilitation of geographical distribution;
- 6) support for heterogeneity and interoperability;
- 7) real-time application response.

3) *Multiaccess EC*: MEC, as highlighted in Fig. 2(c) and discussed in [20], represents a standardized approach developed by European Telecommunications Standards Institute (ETSI) [21], [22] to address the requirements of critical and real-time application needs in 5G and beyond mobile networks. It enables the offloading and data storage tasks from mobile and IoT devices to the edge nodes (edge servers), typically co-located with RAN base stations. Instead of sending all data and computations to central data centers or handling them locally, MEC facilitates efficient processing at the network's edge, enhancing performance and reducing latency.

Opposite to fog nodes, MEC servers can take advantage of the current information on the RAN resources and their statuses. Together with another RAN service, network slicing [23], [24], MEC improves wireless use cases that need low latency, such as AR, VR, and extremely responsive IoT applications. This type of architectural awareness and support is missing in typical FC architectures. MEC architecture provides an additional tier in the RAN between IoT nodes and the cloud, which provides real-time data processing and low-latency demands, and deals with huge amounts of IoT-produced data as well. An example of MEC and its role in 5G networks is studied in [25].

4) *Local EC*: The local edge [Fig. 2(d)] or far/extreme edge, depending on the chosen perspective, refers to clusters of local nodes capable of running lightweight virtualized tasks, located in local, personal, or body area networks [26], [27]. End-user devices, embedded computing nodes, and IoT sensors, actuator, or IoT gateway nodes are common places to find local EC instances. Local EC allows for real-time data processing at the network's edge. Unlike CC, FC, and MEC, local edge nodes can still work together locally even if there is no active connection to the rest of the Internet architecture. It can potentially satisfy many IoT requirements, e.g., low latency, high bandwidth, reliability, energy efficiency, etc., as the data sources and the associated decision-making logic or services are placed at nearby computing nodes [26].

However, IoT sensors and actuators are constrained in nature with limited computational capacity and may operate in extreme cases with a coin-size battery, depending on the use-case scenario [28], limiting their applicability for heavier computing. Despite these limitations, local EC is extremely useful for many IoT scenarios, especially those requiring high resilience for network problems and low latencies, while inflicting low resource consumption for the constrained IoT sensors and actuators to preserve their battery life.

5) *Three-Tier Edge-Cloud Continuum*: With a three-tier or full computing continuum [Fig. 2(e)], we refer to an architecture utilizing the best of the previously introduced models. It comprises three logically connected computational tiers, where tasks can run on any of the three tiers, depending on the situation, either on the local edge, access edge, or data centers [29]. The local EC tier, as described in Section II-A4, operates near the end users, handling immediate processing needs with ultralow latency for lightweight time-sensitive tasks. The MEC tier is situated at the network edge (e.g., at RAN base stations), providing in addition to ultralow-latency-enhanced computational capabilities and serving as an intermediary between the

local edge and the cloud tiers. The CC tier, consisting of centralized servers at data centers, offers extensive processing power and storage for heavy and less latency-critical tasks. In this computing continuum, multitier service orchestration plays a crucial role by dynamically allocating and managing tasks across the three tiers based on factors like resource availability, network conditions, and service requirements, ensuring optimal performance, efficiency, and user experience.

We acknowledge the existing survey literature on traditional IoT-cloud, FC, MEC, local EC, and the three-tier edge-cloud continuum architecture. Recognizing the current research, this work investigates the important gaps and challenges in resource-aware service orchestration, scalability, and efficiency in the EC-CC continuum in 6G and contributes to present the latest perspectives and techniques to improve IoT application orchestration.

B. Related Work

CC infrastructures are typically deployed in large centralized data centers, where computing power, storage, and networking resources are aggregated for scalable and efficient processing. However, scenarios such as massive IoT, autonomous vehicles, healthcare, smart buildings, and smart devices drive extensive decentralization, enabled by the evolution of the EC-FC-CC continuum [3]. Maia et al. [30] investigate the importance of AI and collaboration in next-generation network infrastructures (NGNIs) and highlight the outcomes of the integration of communication, computation, and caching (3C) in the cloud-to-EC continuum. This article explores examples of use cases that demonstrate the requirement for 3C services' emerging resource-intensive applications and explains the system's architectural, legal, and commercial challenges from several angles.

Gkonis et al. [31] reviewed the key developments in the IoT-edge-cloud (IEC) continuum, besides the advancements in subsidiary technologies and their problems. This also highlights the important drawbacks of IoT deployments and computational requirements, which consider potential research directions in IEC continuum systems. Cooperation with the next-generation wireless communication networks presents use-case scenarios that illustrate how this IEC continuum computing node can operate together to provide the best possible service deployment and enhance the user experience. Mishra et al. [32] investigate the key methods of federated learning among emerging technologies such as EC, CC, IoT, Internet of Vehicles (IoV), and vehicular EC (VEC) with local model updates and cloud-based integration.

Golpayegani et al. [33] study the requirements of adaptive EC for the IoT applications operating with a distributed computing paradigm that connects the actual edge devices to data centers via EC nodes. The distributed computing paradigm focused on applications like industrial IoT, smart transportation, and crucial physical healthcare monitoring, where unpredictable environmental changes affect the system performance and decision-making process. However, additionally, it evaluates the modern adaptive EC systems for vibrant environment perspectives. Liang et al. [20] conduct a comprehensive

review of MEC, including the architecture, its applications and service requirements, and standardizing approaches. It also investigates the supporting technologies such as software-defined networking (SDN), NFV, information-centric networking (ICN), service function chaining (SFC), C-RAN, FC-based RAN (F-RAN), and network slicing.

Yang et al. [34] emphasize the evolving security and privacy challenges in 6G networks, particularly due to the integration of advanced technologies like IoT, EC, and AI. These studies highlight how security risks vary across different tiers of the network, given the diversity in computational capacity and threat exposure. A key takeaway is the need for adaptive security strategies [35] that balance stringent protection with resource constraints at the edge. Moreover, Gkonis et al. [31] explored the IEC continuum, focusing on security orchestration and resource management across different tiers. This article highlights how distributing security measures and optimizing computational tasks can help mitigate the challenges posed by resource-constrained devices at the edge. Efficient orchestration frameworks ensure a balanced load while maintaining robust security, offering valuable insights into scalable and secure system design for future 6G networks.

Chiang et al. [36] present a comprehensive analysis of EC, comparing the computing architectures such as MEC, FC, and Cloudlets and investigating the management strategies such as resource distribution and task offloading. However, to enable successful resource management and task orchestration, also discuss the IoT deployment scenarios, development tools, and EC paradigm. Haibeh et al. [37] examine service installation and auto-resource scaling in MEC-NFV systems within MEC infrastructure, virtualization, and management of resources. Considering different perspectives of end users, infrastructure stockholders, and service providers, it analyzes the key strategies for VNF placement optimization.

Primarily on a three-tier architecture consisting of the item local, edge, and cloud layers, Luo et al. [38] review existing research on resource scheduling in an EC environment. It explores four collective methodologies for major challenges, including data computational offloading, resource scheduling, resource provisioning, and resource allocation. Li et al. [39] present a comprehensive taxonomy of cloud-fog-edge architecture and review resource management strategies across the cloud, fog, and edge layers. This article classifies the research based on fundamental challenges, such as the need for low latency in applications, energy consumption, resource availability, and scalability, and discusses important heuristic optimization strategies.

An overview of MEC and a summary of its background, main concepts, and distinctions from other computing paradigms are presented in [40]. It investigates MEC system architecture and highlights recent developments in computation offloading methods, including Q-learning and deep Q network (DQN). It also categorizes and contrasts offloading algorithms according to techniques and use cases. Ismail et al. [41] investigate resource scheduling in MEC with an emphasis on reinforcement learning (RL) and deep RL (DRL) approaches. It categorizes studies into content caching, resource management, and computation offloading

while analyzing key challenges and unexplored research areas.

C. Key Contributions of This Article

Several studies on computational resource management, comparing architectural frameworks, and service orchestration approaches across the EC, FC, and CC paradigms have been conducted; however, several research gaps exist in the existing literature. In 6G networks, related research work has mostly supported individual computing tiers and technologies, also deficient in a consistent viewpoint on the orchestration of IoT services over the multitier continuum. However, limited surveys offer a comprehensive investigation of a multitier distributed computing paradigm, such as MEC, FC, EC, and centralized CC, considering the architectural integrations along with the service-specific requirements.

Survey studies in the literature on the topic of IoT service orchestration mostly focus on CC, FC, or EC paradigms independently and separated from other tiers, which restricts their capability to address across-layer coordination, diverse computing requirements of IoT services, and location-aware computation. Identifying these central research gaps, this article contributes to combining existing knowledge into an integrated multitier edge-cloud continuum architecture that supports inventive orchestration of IoT applications through enhanced resource optimization and low latency at the local edge processing.

The integrated emphasis of EC and AI for IoT service orchestration is mainly unexplored even though numerous works discuss AI separately in these multitier computing layers. This article overcomes the gaps by presenting an inclusive review of IoT service orchestration, highlighting the cooperative integration of the computing continuum in the multitier architecture. The possibility of adopting orchestration and highlighting how AI-based technologies can improve distributed computing offered low latency and instant decision-making in 6G-enabled IoT environments.

In this article, we present a review of IoT service orchestration to the edge-cloud continuum in the environment of 6G networks. The key contributions of this study are as follows.

- 1) *Analysis of the Three-Tier EC-CC Continuum:* In this article, we give a constructed explanation of the three-tier computing architecture, encompassing IoT, traditional cloud, EC, FC, and MEC, emphasizing their roles in supporting IoT service orchestration.
- 2) *Review of Existing Survey Articles:* We examine this article on the IoT service orchestration and identify the challenges and strategic research gaps. The analysis in this article provides an overview of the most fitting computing architecture based on the service requirements.
- 3) *IoT Service Orchestration in Distributed Computing Paradigms:*
 - a) *Traditional IoT-to-CC:* We survey the limitations of centralized CC, specifically for communication latency and network traffic for resource-intensive applications.
 - b) *Service Orchestration in FC:* We study the role of FC in the computing continuum for time-critical

IoT services and its benefits, limitations, challenges, and acceptance in distributed computing.

- c) *Service Orchestration in MEC*: We discuss the role of MEC in the low-latency requirement of applications and its integration within the three-tier computing continuum.
- d) *Service Orchestration in Local EC*: We investigate the essentials of computing at the local edge closer to the source of data for time-sensitive applications, highlighting their resource constraints and service deployment challenges.
- e) *Service Orchestration in Edge-Cloud Continuum*: In this article, we presented the integration of CC, FC, MEC, and EC, evaluating the advantages and challenges of this architecture in supporting efficient IoT service orchestration.

- 4) *AI for Edge and Edge for AI*: In this article, we also review the interchange between AI and EC, emphasizing how AI will increase edge-based decision-making. Also, this article describes how EC supports AI-driven IoT service orchestration (see Section IV).

This article appears as a foundational resource for researchers by providing an in-depth understanding of IoT service orchestration in the three-tier edge-cloud continuum and steering future research directions.

III. INFRASTRUCTURE COMPOSITION OF MULTITIER EDGE-CLOUD CONTINUUM IN 6G

The 6G technologies are expected to support communication for the next-generation IoT devices ecosystem and their services by offering HRLLC, mMTC, and ISAC capacities, converging computing, intelligence, and communication at the edge. These advancements in 6G networks support real-time digital twin systems, intelligent EC, and large-scale, robust, and autonomous systems, all aimed at expanding the IoT toward the IoE ecosystem. Contrasting the previous generation networks like 4G and 5G, which were mainly focused on broadband, low latency, and basic machine-type communications, 6G proposed an AI-centric network that incorporates terahertz communications and persistent AI in distributed computing [42].

The 6G network technology primarily improves the interface between the communication infrastructure and IoT service orchestration. However, it aims to enable highly adaptive, distributed computing, and intelligent 6G-IoT systems. For example, federated learning and explainable AI systems have been projected for intelligent 6G-IoT systems and their infrastructure optimization. This also demonstrates that 6G networks with integrated AI support can improve the performance, scalability, and decision-making capacity of the systems for IoT service deployments [43]. Likewise, converging sensing and communication with ISAC and JCAS empowers the system efficiency, for example, concurrent use of the same radio signals and their reflections for data transmission and sensing, supporting Internet of Medical Things (IoMT) healthcare applications such as patient localization, fall detection, and vital-sign monitoring in 6G and beyond networks.

In this section, we first discuss the challenges depicted by traditional CC for IoT service orchestration and the diverse components of CC, and then investigate the rise of FC, its importance, and its contributions to the computing continuum. Next, the essential role of EC and MEC in the IoT service orchestration and the importance of computation at the local edge, near the source of data, is highlighted. The orchestration of resource-aware IoT services across the edge-to-cloud continuum is studied, which also outlines the limitations of traditional CC and highlights the need to integrate the EC paradigm.

A. Orchestration in Traditional IoT-Cloud With 6G Networks

In the last few years, CC has gained widespread acceptance in online applications and services, such as IoT (e.g., smart homes and smart cities), business applications [e.g., e-commerce and enterprise resource planning (ERP)], healthcare (e.g., telemedicine and health monitoring), and shared repositories. CC enables users to access information at any time from any location, thus facilitating seamless accessibility for users of the cloud. Overall, CC offers various advantages to its users, such as backup of data and its recovery in the case of system crash, global access over the Internet, cost efficiency, ease of deployment, software integration, flexible resource capacity, and pay-as-you-go billing. It is important to acknowledge that there are certain limitations associated with CC. It relies heavily on a stable and high-performance network connection, where any disruption can hinder access to services.

Another concern is latency, since data transmission between users and remote data centers can introduce delays, making CC generally unsuitable for real-time applications. Third-party data management inflicts security risks, increasing the potential for unauthorized access and compliance challenges. Furthermore, users and service providers have limited control over cloud infrastructure, limiting customization, performance optimization, and adherence to specific operational requirements. These challenges have driven interest in alternative solutions like EC and hybrid cloud models [44].

Network Connection: CC assumes the user has a dependable network connection. If there are system availability and network connectivity issues, access to the cloud becomes an issue. The presentation of the cloud applications relies upon the exhibition of the system from the users' perspective. Data transfers take longer than a local server's [45]. *Control of Data Security*: In a typical cloud system, users do not have control over their information and data security policies. The user information can be vulnerable to hacking or phishing assaults [46]. *Additional Costs*: Even though CC generally brings cost-efficiency advantages, hidden costs may be added by, e.g., third-party subscription [47].

The components of the CC service model are illustrated in Fig. 3. *Software as a Service (SaaS)*: Software or applications are delivered as a service running on remote servers and are accessed by end users through their devices. However, users typically have limited control over these services, with access restricted to predefined functionalities. They do not have visibility into the underlying infrastructure, and access to these

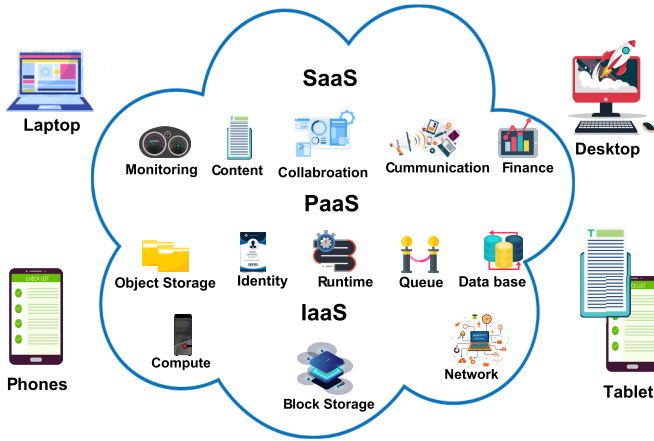


Fig. 3. Components of CC: SaaS, PaaS, and IaaS.

services is subject to limitations set by the provider, as seen in applications like Internet browsers. *PaaS*: A cloud-based environment where developers can build, deploy, and manage applications without handling the underlying infrastructure. Unlike SaaS, which is targeted for end users, PaaS offers tools, frameworks, and runtime environments that allow software developers and service providers to create, customize, and deploy applications while still abstracting away the infrastructure management. *Infrastructure as a Service (IaaS)*: It provides virtualized computing resources such as servers, storage, and networking on demand. Unlike SaaS, which delivers fully developed applications, and PaaS, which offers a development environment, IaaS gives the service administrator direct control over the infrastructure components of the CC environment. This allows scaling resources, like CPU, GPU, and storage capacity, for end users, while managing applications and configurations, making it a flexible solution for hosting, development, and enterprise ICT operations [48].

1) *Limitations in CC and Gap Analysis*: The following are some limitations in CC and the gap analysis in the edge-cloud continuum.

Attacks on the Availability of Cloud Services: As the CC infrastructure is publicly available, the applications deployed on the cloud can be intentionally or unintentionally disrupted by other users or intruders, and users may lose access to their private data. Denial of service (DoS) and distributed DoS (DDoS) attacks are good examples of such attacks [49]. Here, a malicious entity attempts to overload a server by sending request floods from a single source (DoS) or multiple sources (DDoS), thus making a service hosted by the influenced server unavailable for its users.

Dependence of Connections: To access the services of CC, continuous Internet connectivity is required. This is a particular challenge in mobile use cases and when the user is located in a remote area. The Internet connection in both of these cases can be unstable, or its performance can be low in general. In addition, the network connections may be hindered at times by, e.g., network congestion and physical failures.

Low Performance of Connections: The abovementioned network issues may also cause low performance of the network

connections between the end users, data sources, and cloud services. Furthermore, in highly centralized deployments, a high amount of concurrent users may cause network congestion at the server end, leading to low performance.

High Latency: Remote location of end users, congestion, and network failures typically cause low latency, in addition to bandwidth limitations. Furthermore, even good-quality connections are unable to provide low latency if the physical distance of the endpoints is very high due to cumulative signal propagation and switching delays of long routes.

Security and Privacy: Traditional CC environments face persistent risks of unauthorized data disclosure. This stems from the physical separation between cloud servers and data sources, requiring data to travel through public networks. Such transmission increases the attack surface and potentially exposes sensitive data to interception. To mitigate these risks, encryption techniques like Advanced Encryption Standard (AES) and Data Encryption Standard (DES) are utilized. However, these algorithms are computationally heavy and introduce latency and overhead, especially in resource-constrained environments [50].

In the conventional IoT-cloud continuum, the deployment of applications and orchestration of services are well-suited for activities that need resource attention. However, those are not optimal for low-latency applications and services. In the context of CC, it is important to consider that although the cloud may provide enough computation and processing resources, there is a tradeoff in the form of increased latency as illustrated in vertical bars in Fig. 1. The IoT devices can establish connections and communicate with local-edge data centers through different means of communication such as Wi-Fi routers, Bluetooth, and other short-range communication technologies (see Fig. 1). A comparative analysis of orchestration in the traditional IoT-cloud continuum to the other architectural frameworks discussed in this article is presented in Table II. To bring the data computation and processing of IoT services and applications closer to the data source, FC plays an important role in this computing continuum, which is explained in Section III-B.

B. Orchestration in FC With 6G Networks

EC has emerged as a way to address issues related to the abovementioned concerns in CC. With the rapid development of IoT devices and the increasing demand for cloud services, EC enables moving data processing closer to the end users and the sources of data at the edge of the network rather than relying solely on centralized cloud infrastructure. The widespread adoption of IoT requires instant and secure computation, where many edge devices will process sensed data locally with virtualized components part of the cloud system, reducing the need to transmit it to the cloud, and enabling more applications to operate efficiently at the network edge [51]. EC, presented in Section III-E, is a localized version of CC that distributes data computation and storage to edge devices instead of a central data center. In this case, only the results of the data processing need to be transported over networks. In this situation, EC architecture provides precise results, offers

TABLE II
KEY ASPECTS OF ARCHITECTURAL FRAMEWORKS IN THIS ARTICLE

Category/Aspect	Traditional IoT-cloud	Fog Computing	MEC	Local edge computing	Three-tier computing continuum	References
Example Use Cases	Traditional Smart home	Industrial IoT	Mobile real-time applications	Local lightweight Real-time Applications	Autonomous traffic, Computer-assisted surgery	[61], [59]
System / Generic Requirements						
Scalability	Limited	Medium	High	Low	Very High	[62], [63]
Reliability	Low	High	Very High	Very High	Very High	[57], [64]
Mobility Support	Limited	Medium	Very High	Low	High	[60], [65]
Cost	Low	Low to Medium	High	Low	Medium	[66], [67]
Local Device Requirement	Minimal	Medium	Medium to High	High	Flexible	[68]
Computation						
Data Processing	cloud nodes	edge and cloud nodes	edge and cloud nodes	local-edge nodes	Local, edge and cloud nodes	[3], [6]
Data Storage	cloud storage	Local and cloud storage	edge and cloud storage	Local storage	Local, edge and cloud storage	[18]
Resource Ownership	Centralized	Distributed	Organization-owned	User-owned	Distributed	[66]
Resource allocation	Centralized	Distributed	Distributed	Local	Distributed	[69], [41]
Energy Efficiency	Low to Medium	High	very High	Very High	Very High	[70]
Communication						
Bandwidth Utilization	High	Low to Medium	Low to Medium	Low	Low to Medium	[71]
Network Latency	High	Low to Medium	Very Low	Ultra-Low	Ultra-Low	[72]
Others						
Exposure to Security threats	High	Medium to High	Low to High	Low	Low to High	[73]
Security Capabilities	High	High	High	Low	High	[74]
Network Slicing Support	Limited	Limited	High	Low	Moderate to High	[75]
Service Orchestration	Support	Support	Support	Support	Support	[76]
Service Deployment Location	cloud nodes	fog nodes	cloud and edge nodes	local-edge nodes	cloud, edge and local-edge nodes	[77]
Management Complexity	Low	Medium	High	Low	High	[56]
Standardization	IEEE, IETF	IEEE, IETF	ETSI, 3GPP	Custom	ETSI, 3GPP, ONF	[78]

low latency, and consumes far less network bandwidth than in conventional CC [52].

The integration of EC with FC enables the scalable and resilient deployment of IoT applications on 5G and beyond networks, satisfying the stringent requirements for network slicing, latency, and system reliability within the edge-cloud continuum. In [44] and [53], Cisco introduced the novel notion of FC that extends CC and its services toward the edge of the network. FC provides a bridge between edge devices and CC data centers for operations such as AR, content delivery, caching, big data analysis, data computation, and network services that require low latency, high throughput, and high reliability [53]. The FC layer is positioned on top of the edge to integrate smart devices, switches, and routers with high computing capabilities [54], [55].

Elements of FC are apparent in the smart grid and connected vehicles, when a user at a shopping center is more likely to be interested in local sales, or when a visitor to a park could be curious about the local weather [51]. The concept of FC requires location awareness, improved QoS for real-time streaming, low latency, and real-time applications (automation, transportation, sensors, and actuators of the network) [56]. Fog devices are mainly end-user devices, access points, routers, and switches. Fog devices serve as centralized coordinators between cloud and edge (autonomous vehicles, sensors, actuators, etc.) in a distributed manner [57], [58]. The fundamental concepts of both cloud and fog in the network

include computing, storage, and networking resources. FC has many distinct characteristics, including mobility, extensive geographical distribution, low latency, location awareness, a substantial number of nodes, real-time streaming capabilities for durable applications, wireless accessibility, and heterogeneity [59], [60].

1) *Rise of FC*: The fundamental goal of introducing FC was to outspread services and functionalities offered by the cloud at the edge of the network [18]. For instance, fog devices can be installed inside airports, train stations, bus stations (even inside buses), shopping centers, and public parks. By this, users at the edge can benefit from the high bandwidth local connections with low latency and power [51]. The applications of FC precached the information of the frequent positions of the environment and responded to the queries asked by the IoT and edge devices. Communications systems such as vehicle-to-vehicle, vehicle-to-infrastructure, and machine-to-machine, with FC devices strategically deployed in shopping centers and public parks, are widely acknowledged as notable examples of FC and EC [18], [51]. Application developers can access, write, and update sensor configurations on automobiles and machinery by adopting these frameworks and interacting directly with these resources [61], [79].

Cloud-to-edge services face latency, resource management, load balancing, and connectivity challenges, and FC is useful for solving these problems [64]. The main concept of FC is to provide a decentralized data computation system for the

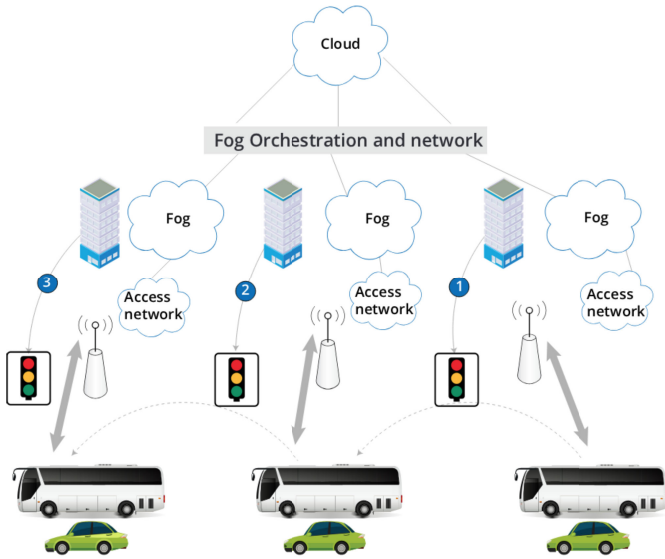


Fig. 4. Smart traffic lights and connected vehicles.

sensors and devices connected at the edge of the network [53]. As shown in Fig. 4, smart street traffic lights networked locally with sensors and actuators, detect the presence of pedestrians and bikers alongside the road, and measure the distance and speed of forthcoming vehicles and support their mobility [60], [65], [80].

2) *Contribution of FC Into Edge-Cloud Continuum*: One of the methods that can satisfy the demands of an increasing number of connected devices and IoT applications is FC [68]. It uses the local resources at the edge rather than remote computer resources, increasing performance and reducing bandwidth issues. Fog and CC are interconnected [81], and this section will provide scenarios of FC that determine CC and impact performance.

- 1) *Smart Grid*: FC is software that executes on network devices in smart grids. It provides consistent machine-to-machine (M2M) communication between connected devices and data-processing services without sending queries to the cloud. Load balancing, task scheduling, resource management, and energy-efficient distribution of applications will possibly run on devices at the edge, like smart meters and microgrids [19], [70]. Due to the shortage of energy and its high demand, accessibility, and the lowest price, these network edge devices can automatically switch to substitute energy resources like solar and wind [80].
- 2) *Smart Traffic Lights and Connected Vehicles*: FC paradigms can play a vital role in the field of automobiles, traffic cameras, sensors, and actuators installed in vehicles. Traffic lights and cameras on the poles sense an ambulance by its blinking lights and can automatically switch street traffic lights to allow the ambulance and police cars and open lanes for vehicle traffic [80].
- 3) *Wireless Sensor and Actuator Networks*: Conventional networks of sensors may not be fully functional in evolving applications that turn out to sense and track the situations in the environment, but these necessitate the actuators to employ actual actions like opening

and closing, or even resonant sensors. The connectivity between the continuum of edge, fog, and cloud in hierarchy order can be a plausible appreciation to several system innovations, including wired and remote technologies [9].

- 4) *Fog Connectivity*: Wireless connections may ordinarily link the infrastructure of IoT devices to the fog nodes since these devices regularly have only remote interfaces [72]. Wireless connections can likewise be utilized in fog-to-fog or fog-to-cloud interconnections, relying upon the accessible framework. Cellular technologies (3G, 4G, beyond 5G, and 6G) are expected to be used in FC paradigms [54].
- 5) *Security and Privacy in FC*: Unlike cloud-based data centers, devices in the FC paradigm are spread geographically and consist of heterogeneous platforms [82]. Additional security capabilities, such as distribution and revocation of certificates, can be configured at the fog layer [83], resulting in slightly improved security capabilities as compared to CC-based data centers. The threat exposure remains approximately the same. The transition from CC to FC makes the security risk similar to the addition of new security capabilities.

The key influence feature involved in the success of the edge-fog-cloud continuum of computing is the elasticity and scalability of the resources [63], [84], [85]. Typical targets and prerequisites of CC applications are adaptability, mobility, flexibility, and effectiveness (i.e., scalability, elasticity, and efficiency). These quality attributes are not well-standardized, and a large number of classifications, explanations, and measurements are accessible in this article [86]. There are divergent perspectives among researchers about FC. Although it is positioned between the first and third tiers of the computing continuum (Fig. 1), FC has the potential to be used in industrial and location-based IoT application deployment to provide both high computational capabilities and low latency requirements [77]. This article provides a comparative analysis of orchestration in FC concerning the alternative architectural framework of MEC, described in detail in Section III-C. The findings of this analysis are shown in Table II.

C. Orchestration in MEC With 6G Networks

In the context of advanced mobile network generations like 5G and Beyond, MEC plays an important role in meeting the requirements of time-sensitive applications operating in real-time environments, ranging from AR and autonomous vehicles to remote healthcare surgeries [87]. A cloud server inherently possesses substantial storage capacity and powerful computational capabilities. Furthermore, the centralized model offers an ideal environment for data processing. Nevertheless, advanced technologies demand high data speeds and can be constrained by latency limitations. As a result, distant data centers may encounter challenges within the aforementioned constraints. Therefore, MEC servers, which enable EC, emerge as prime candidates for real-time data processing and applications where low latency is critical [88].

In the MEC-cloud continuum, orchestration relies heavily on resource allocation and collaborative task offloading [69]. The effectiveness of task offloading is intricately tied to

the hardware and software specifications of computational nodes, which determine their capacity to host specific tasks efficiently. Huang et al. [89] investigated cloud-MEC-based collaboration computing architecture and proposed a collaborative task offloading scheme with service orchestration (CTOSO) to address the demands of emerging applications by considering low latency, high complexity, and enhanced reliability. The proposed scheme addresses the main issues of task offloading and specific offloading for terminal selection, and it also improves the network performance efficiently. The experimental results in this article demonstrate that the scheme improves latency performance as well as ensures efficient energy consumption.

Guo and Liu [90] explore collaborative computation offloading in hybrid fiber-wireless (FiWi) access networks and introduce solutions to reduce computation time and energy consumption. These solutions perform better than noncloud alternatives, especially for more complex computational tasks. Over time, domains such as the IoMT, automotive technology, and digital healthcare have experienced a growing number of time-sensitive use cases that require increased computational capacity and ultralow latency. Silvestro et al. [91] introduced a model called Mute (multitier edge networks), which outlines multitier edge nodes and cloud architecture for task offloading. This model strategically places services at specific nodes on the edge servers, resulting in a substantial reduction in both latency and task completion time within the architecture.

The task offloading in a three-tier distributed edge-cloud continuum architecture can vary based on the computing hardware properties, communication protocols, and use cases. The increasing quantity of IoT devices and the necessity of delegating their responsibilities to optimize resource use are covered by Almutairi and Aldossary [92]. This article discusses variables like CPU consumption, network demand, and delay sensitivity, and presents a novel method for optimizing task offloading in an edge-cloud system utilizing fuzzy logic algorithms. The edge tier outperforms the local tier due to the presence of MEC servers, which provide a lightweight version of cloud data center capabilities closer to the network edge. Although the centralized cloud tier remains a key component thanks to its known strengths, use cases demanding low latency, encountering network congestion, or involving locally generated data without broad distribution emphasize the need for a scalable architecture, which can increase the utilization of local and edge tiers, supported by cloud data centers [93], [94].

The flexibility of computational and storage services near the user at the edge makes it up against an overwhelming number of highly sophisticated security threats [74] as data are spread across different nodes and storage devices, including servers, desktop workstations, mobile devices, and wireless network sensors. The interest of research lies in the trustworthiness of the MEC environment. To ensure interoperability among the various layers of the MEC framework [22], security management is essential. These layers include communication technology (CT) capabilities, IT applications, the MEC platform, devices, and the edge-cloud. MEC security architecture [95] introduces packet inspection, malware detection, traffic

monitoring, and security policy enforcement into SDN-based centralized control at MEC. MEC nodes are susceptible to new types of security threats such as eavesdropping, information injection from rogue nodes [74], and side-channel attacks.

D. Orchestration in Local EC With 6G Networks

Cisco¹ and Ericsson² estimated that more than 29.3 billion connected devices were actively working at the end of 2023, and more than 2.5 billion IoT devices will be operating by 2027. Collectively, these devices are producing a zettabyte of data that has resource and network requirements for computation and communication, respectively. The current two-tier service deployment architecture, CC and EC, is hardly handling the increasing number of requests coming from these IoT devices. In some cases, due to changes in the requirements of the IoT sensors and actuators, services that are already deployed at the CC and MEC servers are being offloaded [29].

Earlier, service deployments were typically conducted using multiple virtualized server components such as CPUs, GPUs, memory, and storage by leveraging virtual machines (VMs), rather than provisioning new physical servers. Hypervisors (e.g., Oracle VirtualBox and VMware) enabled each VM to run its own operating system (OS), thereby providing isolation between system components. However, researchers have identified several limitations of VM-based deployment, including inefficient resource utilization and lengthy boot times [96]. Moreover, system engineers and administrators were often required to develop custom algorithms to handle load balancing and scalability across VM clusters, a task that is both complex and prone to errors. As a result, VM-based orchestration has proven to be inadequate, particularly in CC and MEC contexts, prompting the search for more suitable technologies to support the demands of heterogeneous IoT infrastructures.

In the container-based application deployment, a Docker image uses the kernel that provides bridging between hardware and software applications from the host OS. But in VM-based service deployment, it uses a separate dedicated kernel based on the OS from where the VM is operating [96]. Compared to VM-based deployment, a container only loads the host kernel rather than loading a full OS to share hardware resources, i.e., CPU, GPU, RAM, and storage. As a result, a container works like a single process, which is very lightweight as compared to a VM. Kubernetes, Apache Mesos, and Docker Swarm are the most popular container orchestration platforms.

Lightweight IoT computing nodes (e.g., RPis and Nvidia Jetson Xaviers) include sufficient processing power to process data and execute tasks locally based on the decision-making services [27]. To ensure the overall QoS of an IoT network, Harjula et al. [29] presented in their *nanoEdge model*-based proof of concepts (PoCs), how services can be deployed locally to perform actions and reduce the excessive load at EC and MEC. In their PoCs, each service includes a

¹<https://www.cisco.com/c/en/us/solutions/collateral/executive-perspectives/annual-internet-report/white-paper-c11-741490.html>

²<https://www.ericsson.com/49d3a0/assets/local/reports-papers/mobility-report/documents/2022/ericsson-mobility-report-june-2022.pdf>

granular functionality of reading sensor data, storing it in a database in the form of nanoservices rather than monolith or microservices, which were used at CC and MEC. However, Docker and Docker Swarm are used to deploy and orchestrate the containers across the cluster nodes [28].

Computing at the local edge, as shown in Fig. 2(d), is considered comparatively secure since there is no communication to the MEC or to the cloud, and the entire data processing and communication happen locally. The threat spectrum is low as the exposure to potential hostile nodes is minimal. While the computational resources are low, they can be minimally occupied for the provision of security since the attack surface is limited and in control. The local tier of three-tier edge–cloud continuum architecture is more resource constraint in terms of computational capacity and storage, whereas this tier can host lightweight tasks, however, as well as some computationally demanding tasks of medical imaging such as cone beam computed tomography (CBCT) image reconstruction and breast cancer analysis by leveraging GPU-supported local hardware, as it presented in [27].

Shahid et al. [97] have briefly presented deployment results of IoT service orchestration in edge–cloud continuum using swarm intelligence-based AI-driven algorithms with ML approaches. The results demonstrate that the proposed algorithms are significant and outperform the baseline solution in terms of maximum application success rate, less energy consumption, resource optimization, and load balancing in the computing nodes. The IoT services that require minimal computational resources but demand instant response time will be orchestrated to the local computing nodes at tier 1. Likewise, AI-driven and ML-based orchestration algorithms perform in a way that the IoT services that demand the optimum level of computing resources and need low latency will deploy to the EC nodes at tier 2.

IoT devices are resource-constrained and mostly are battery-powered with limited resources. Deploying uniform security methods across the edge–cloud continuum may lead to energy exhaustion. Security capabilities are spread across the tiers, responding to attacks from potential hostile nodes at any tier. Adaptive security is imperative for 6G in the edge–cloud continuum, being able to assess the conditions and respond to arising threats. Adaptive security models are designed to dynamically adjust security solutions in response to changing network conditions and evolving threats. Existing approaches such as risk-based [98], game-based [99], requirements-driven [100], event-driven [101], ontology-based [102], and adaptive security models will be evaluated for edge–cloud continuum considering their applicability and adaptability.

E. Orchestration in Edge–Cloud Continuum With 6G Networks

Increasing data computing capacities in edge devices, IoT, and sensors connected at the edge of the network demand local computation of data to avoid the latency cost and entirely exploit scalable data computing at the edges of the network [66], [67]. While the integration of EC and the cloud holds significant potential and attraction, it is crucial to acknowledge

the existing limitations at the edge side that require close attention of researchers and developers [103].

In a distributed multitier edge–cloud continuum environment within 6G networks, the deployment of IoT applications and services expects intelligent management of both computing and network resources. These IoT applications are technically composed of microservices that require a diverse set of resource requirements like CPU, memory, storage, latency, and bandwidth. The integration of EC and FC allows localized data processing, reduces the reliance on centralized CC infrastructure that supports ultralow latency and high data rate. Advanced intelligent orchestration methods for IoT applications dynamically allocate computing resources within EC, FC, and CC tiers as in Fig. 1, considering a complete computing continuum, based on service-level agreements (SLAs), application requirements, and network availability [30].

Adopting advanced 6G network features like network slicing, AI-driven resource management, and URLLC, an intelligent orchestrator can ensure optimal service deployment and QoS [104]. The edge–cloud continuum breaks down the centralized data center barriers and offers novel forms of computing where computing power and data resources are gradually accessible from anywhere and anytime. Orchestration of IoT applications in edge–cloud continuum is illustrated in Fig. 2 and explained in Section II-A.5. Section III-E.1 explains how EC forefronts to give solutions to the problems faced in traditional CC.

1) *From Cloud to Edge Motivations for Decentralized Edge–Cloud Continuum Architectures*: The ultimate objective for the next-generation CC systems is to solve problems and challenges like latency and single point of failure by multiple cloud providers and offer a decentralized computing paradigm nearer to end users. Wang et al. [71] discuss resource management, task scheduling, service placement, and load-balancing challenges while relocating between standard CC and decentralized edge–cloud continuum. Resource management refers to the methodology employed in the allocation of storage capacity, computational power, and network resource optimization by semantic network slicing to a diverse range of applications [75], [105]. Service or container orchestration based on the resource requirements is a significant deployment method for resource optimization in edge–cloud continuum nodes [76]. In resource management, the primary objective is to effectively fulfill the requirements of various applications while prioritizing the satisfaction of user demands [55], [106].

Orchestration of IoT application in the edge–cloud continuum with 6G takes advantage of both EC and traditional CC, while it benefits from edge devices where proximity to the endpoints [107]. The fundamental use case here requires bringing intelligence close to the edge, where excessively low latency is critical and is assured by FC. Shahid and Pahl [108] presented the enhanced particle swarm optimization (E-PSO) algorithm to obtain the multiobjective optimizations such as effective resource management, task scheduling, load balancing, and particularly how the backup VMs become active while running out of resources in edge–cloud clusters.

Fig. 1 illustrates the three main components of the computing continuum, namely, the IoT, EC/MEC/FC, and CC.

TABLE III

EDGE-ENABLED AI ENCOMPASSES TWO KEY ASPECTS: EDGE AI FOR EDGE AND EDGE FOR AI. THE TABLE PRESENTS RESEARCH ARTICLES COVERING BOTH PERSPECTIVES, INCLUDING A SHORT SUMMARY OF EACH ARTICLE, ITS USE CASE, AND ITS RELEVANCE TO THREE-TIER EC-CC

	Year	Ref.	Usecase	Short Summary	Tier 1	Tier 2	Tier 3
Edge for AI	2020	[111]	Healthcare	The paper investigates the decentralized model training to deal with privacy and security concerns for digital health.	✓	-	✓
	2021	[112]	Networking	The survey paper presents crucial challenges in FL and provides various schemes that categorize FL approaches.	✓	-	✓
	2021	[118]	VK & AD	The article presents the different FL use cases and investigates resource optimization approaches in FL.	✓	✓	-
	2025	[130]	Healthcare	The paper investigates technical requirements of medical imaging use cases in three-tier edge cloud architecture.	✓	✓	-
	2022	[119]	IoT	The survey article focuses on FL and SL distributed learning paradigms and the advantage of combining them in edge-based IoT.	-	✓	✓
AI for Edge	2023	[128]	IoT	The article proposes a task priority algorithm using DRL for task offloading in cloud and edge servers.	-	✓	✓
	2022	[129]	IoT	The research concentrates on DRL and Q-learning, to enhance task offloading decisions to tackle issues for storage management, power consumption, and latency to reduce system costs and effectively manage high-dimensional data.	-	✓	-
	2023	[126]	IoT, IoMT	The paper surveys of distributed AI in EECC, covering key concepts, learning methods, optimization techniques, and computing paradigms. It studies EECC's role in AI, security challenges, defense mechanisms, applications, and open research issues.	✓	✓	✓

TABLE IV

LESSON LEARNED: KEY FEATURES AND ASPECTS OF SERVICE ORCHESTRATION

Feature/Aspect	Traditional IoT-cloud	Fog computing	MEC	Local edge computing	Three-tier computing continuum
Key features	Ensure effective, scalable resource availability for extensive IoT data.	Data processing proximity to user, with low latency communications.	Processing, storage and data computation to the edge of the network.	Enhance data privacy and security. Decision making and processing data locally.	Integrated computing continuum, and enhance QoS performance.
Strong aspects	Well-recognized tools and frameworks for QoS requirements of IoT.	Instant response to time-critical applications, with scalable resources.	Support high bandwidth and ultra-low latency for IoT applications.	Instant data processing for real-time applications, control over data.	Distribute workload in three-tier, ensuring continued application performance.
Weak aspects	Challenges of high latency and connectivity for time-critical applications.	Resource management and security challenges of fog nodes.	Limited resources, challenges of defining orchestration protocols.	Challenges of limited computing resources for massive IoT.	Increased complexity, challenges in adaptive offloading.

EC nodes and conventional CC are outperformed in terms of computational capacity. The MEC and EC nodes are positioned at tier 2 in Fig. 1. In the context of this three-tier architecture, the computational resources available on MEC and edge nodes exceed the capabilities of local-edge nodes. Subsequently, these MEC and edge nodes can accommodate applications that need intermediate computing resources and demand low latency. Table II illustrates the comparison of orchestration in EC to different architectural frameworks in this article.

Since the concept of 6G is still shaping up, there is no consensus on KPIs for the edge-cloud continuum with 6G networks. However, based on the prospective aims and expectations for 6G, some predicted evolution from 5G KPIs and for the five different computing continuum architectures we discussed above are latency, high data throughput, security, robustness, and resilience [109].

The 6G devices and network elements are exposed to various security threats and malicious attacks. If not handled properly, these can pave the way for intruders into the network, thus affecting the security and privacy of the sensitive data in the network. The security requirements, threats, and solutions are different in different tiers as the computational capacity and threat actors vary across the tiers.

IV. EDGE-ENABLED AI EXAMPLES IN 6G

This section explores the potential of distributed AI solutions within a three-tier architecture. We begin by examining the requirements of AI-based applications, focusing on their optimal placement within the architecture while highlighting the computing capabilities of each tier. Next, we discuss AI-driven attributes from both the edge for AI and AI for edge perspectives, including agentic AI approaches for IoT service orchestration, along with a detailed analysis of these concepts. We also present relevant studies from the literature in Table III, offering insights into existing research in this domain.

A. Placement of AI-Based Applications in the Local Edge to Cloud Continuum

The idea behind using AI and deploying AI-based algorithms in the local edge and cloud continuum is to increase the performance and efficiency of the system. It includes making strategic decisions about where to run AI applications while balancing several factors for more efficient AI deployments, such as latency, computational resources, and data privacy. One of the subsets of ML and DL algorithms can comprise many hidden layers that require more computational power when training deep models. To accelerate the training time and

decision-making process, GPU-based computational hardware is a priority for DL-based tasks, and the execution time for deep neural network (DNN) is lower than that of CPUs [110].

Tier 1 of the distributed three-tier continuum architecture in Fig. 1 comprises a variety of local computing nodes. Still, these nodes are resource-constrained in terms of hardware specifications and are mostly capable of supporting lightweight computational tasks. Collaboration among these local nodes and other tiers in the architecture can be leveraged to deploy local AI models. AI-based distributed learning paradigms including federated learning in digital healthcare [111], wireless networking systems [112], and split learning are among the techniques for the edge–cloud continuum.

EC technology enables CC capabilities that ensure the deployment of edge AI algorithms as edge AI models in a two-tier or three-tier architecture. MEC server-enabled edge networks can host computationally demanding tasks, which local nodes struggle to deal with for processing the data. The three-tier computing architecture contains cloud data centers, which provide conventional environments for running/training heavy DL tasks and deploying cloud-AI models. Deep neural networks outperformed traditional image processing algorithms and other ML algorithms in computer vision and image classification [113]. As a use-case example from digital healthcare for medical imaging scenarios, AI and ML algorithms can be leveraged on the EC–CC architecture for diagnostics and disease analysis, such as in mammography [27], dermatology [114], and COVID-19 pandemic [115]. The local EC hardware technical specifications play an important role in deploying AI-based diagnostics.

B. AI-Driven Attributes in Local Edge to Cloud Continuum

The use of AI in the local edge–cloud continuum can be viewed from two different aspects. The first aspect is that the GPU-enabled distributed computing architecture can host the deployment of AI and ML applications by providing computational resources, where edge devices can make independent decisions fast without connection to remote servers, which is known as edge for AI; on the other hand, AI and ML methods can be used to optimize of such architectures to help decision-making, such as in which tier/node the data processing should be performed, to provide efficient resource and data management, which is known as AI for edge [116]. The hardware characteristics of the nodes in the architecture are crucial for a GPU-based system that accelerates the training of DL models. The AI methods that can be used in the distributed EC–CC architecture for both aspects will be detailed in the following.

1) *Edge for AI*: The computational hardware in architecture hosts AI algorithms to train models on datasets in three tiers to analyze data close to the data sources in local edge nodes and remote data servers. The appropriate ML algorithms for this scheme include supervised learning and unsupervised learning such as classical ML- and DL-based algorithms [117]. Federated and split learning are distributed learning paradigms that enable the deployment of ML/DL algorithms that provide decentralized model training. These distributed learning paradigms are applied to EC platforms, as seen in [118], where the article presents virtual keyboard (VK) and autonomous

driving (AD) as use cases [119]. Additionally, edge nodes can leverage pretrained AI models for rapid data analytics. For example, AI models trained on large datasets in the cloud can be used for inference at the resource-constrained edge nodes. Therefore, edge-enabled inference can play a crucial role in delay-critical healthcare applications. The review article in [120] presents a comprehensive definition of edge inference at the network's edge. Furthermore, the study presented in [27] serves as a representative example of the inference of the AI model on GPU-enabled local edge devices. It leverages pretrained DL models for medical imaging data analytics.

2) *AI for Edge*: The heterogeneity in distributed computing platforms brings some challenges in terms of resource allocation, task offloading, and service placement [121]. The optimization of the computing platform is crucial for the improved efficiency of the systems. Since the architecture has a dynamic environment and is subject to change frequently, RL is one of the optimization AI approaches that suit distributed heterogeneous computing platforms. The algorithm interacts with the states in the environment and takes actions based on feedback from the environment [117]. Q-learning is a type of RL based on the Markov decision process (MDP) [122], [123].

Edge-AI roles in IoT service orchestration are important as it enables intelligent resource optimization, decision-making, and dynamic service placement for efficient processing of massive data generated by IoT devices. AI-driven orchestration mechanisms can learn from the system states and workload patterns to manage heterogeneous edge resources effectively, thereby improving overall system performance in distributed computing architectures [124]. Furthermore, AI-based multiobjective RL approaches facilitate efficient utilization of computing and networking resources across diverse edge devices in the edge–cloud continuum, which enhanced energy efficiency for IoT with large-scale deployments [125]. Integrating AI-based swarm optimization algorithms into an intelligent orchestrator is a well-established method for deploying energy-efficient IoT applications based on their resource requirements and the capacity of available resources in computer nodes [104].

Duan et al. [126] present a comprehensive overview of distributed AI in end-edge–CC (EECC), focusing on key concepts, distributed learning algorithms, and computing paradigms. It explores the role of EECC for AI, and also optimization technologies for training and inference. In addition, it discusses privacy and security challenges and reviews defense mechanisms, alongside highlighting promising applications such as smart healthcare, smart industry, real-time video analytics, and open research issues in the field.

IoT service deployment in 6G and beyond networks is progressively intersecting with ISAC and JCAS. By this, the service orchestration frameworks mutually manage the service communication, data sensing, and computational resources to enable high data throughput, URLLC, and supporting high user mobility for IoT services across the edge–cloud continuum. Scaling the ISAC and JCAS capabilities, modern 6G-enabled IoT service orchestrators can vigorously adapt service deployment, resource allocation, and network slicing based on real-time sensing and network conditions. This

results in enhancing the overall system efficiency, intelligence, and scalability for emerging IoT applications [127]. RL plays an important role by providing an intelligent distributed computing architecture. Its application includes from resource allocation to task offloading. Tang et al. [128], utilized DRL for task offloading on a collaborative EC-CC platform. Another example of using DRL for computational task offloading is studied in [129].

The semantic service orchestration that is based on the agentic AI approach proposes to move service management toward meaning-aware coordination [131]. As orchestration decisions increasingly emerge from distributed AI-driven reasoning, ensuring trustworthiness becomes a central scientific challenge, as distributed and multiagent AI systems introduce main vulnerabilities regarding privacy and security compared to centralized solutions. In some domains like healthcare, orchestration must follow strict requirements, including the privacy of confidential patient data and regulations. It is also important to remain robust to the external security threats and system failures across heterogeneous and dynamic edge-cloud architecture. When AI participates directly in distributed orchestration and control decisions, traditional approaches become insufficient, as existing trust and governance tools for AI were developed considering centralized approaches and might not be ready to scale to autonomous, distributed, and adaptive AI-based control loops [132].

V. SUMMARY, LESSONS LEARNED, AND FUTURE DIRECTIONS

A. Summary

This article provides an analysis of this article on five significant architectural frameworks for the study of 6G networks. These frameworks consist of traditional IoT-CC, FC, MEC, local EC, and EC. This article distinguishes between these frameworks based on KPIs relating to service orchestration, resource optimization, IoT application deployment, latency, data rates, security and privacy, robustness, and resilience. It provides an in-depth review of each framework, highlighting its individual strengths and weaknesses. Furthermore, it investigates how each framework facilitates efficient service orchestration, implements IoT applications, controls latency, utilizes resources effectively, and improves data rates, performance, and robustness against failures and security threats in the emerging 6G environment.

This article provides important insights into the impact of different architectural frameworks on the development of edge-cloud systems in the context of 6G. This article also emphasizes the unique features and contributions of each framework. This is particularly significant given the growing demand for robust systems offering low latency, high data rates, and resource efficiency. Evaluating the five frameworks emphasizes the importance of edge-cloud continuum models that cater to the specific needs of 6G networks. However, this article's discussion highlights the need for further investigation and improvements to these architectural frameworks in order to utilize their capabilities effectively in 6G networks.

B. Lessons Learned

Orchestration of IoT applications and services with various computing and communication needs must be assigned to appropriate computing nodes. However, the orchestrator will schedule services so that, for example, time-critical services are deployed to local edge nodes at tier 1; services requiring low latency and sufficient computation are considered for deployment in EC, FC, and MEC at tier 2; and delay-tolerant, high-computation services are directed to the cloud data center at tier 3. The following is a concise analysis of the five computing continuums discussed in this article and presents in Table IV.

1) Service Orchestration in Traditional IoT-Cloud: Substantial orchestration solutions that systematize the deployment of IoT applications and services across centralized CC are a significant sign of the development of traditional IoT-CC. To operate the extensive amounts of data produced by IoT devices, traditional IoT-CC systems architecture ensures effective resource availability and scalability.

Focused Approaches: The primary importance has been focused on establishing open-source, commercial, and adaptive orchestration technologies, such as Amazon Web Services (AWS), OpenStack HEAT, and Kubernetes, which empower cloud resource allocation and system integration.

Strong Aspects: In the CC environment, well-recognized tools and frameworks with extensive support for infrastructure integration and determined QoS requirements, such as reliability and scalability, facilitate the orchestration of IoT applications and services.

Weak Aspects: Considering the service orchestration directly to the CC, it is not suitable for time-critical applications like healthcare. Offloading data to the cloud experiences high latency due to centralized computing, and there is a possibility of network breakdown while sending sensitive and huge amounts of data to the cloud.

2) Service Orchestration in FC: Significant developments are establishing an intermediary computing layer between cloud data centers and end-user IoT devices. FC offers data-processing proximity to data sources, and indeed, adopting FC reduces the network load and latency.

Strong Aspects: FC environment offers instant response time to end-user service requests with low latency. However, it enhances data scalability and resource optimization by adopting a decentralized resource management approach.

Weak Aspects: Resource management of diverse and distributed computing nodes becomes more difficult and also raises the privacy and security concerns of FC nodes.

3) Service Orchestration in MEC: Important developments for IoT applications and corresponding services require ultralow latency and high bandwidth, which is supported by MEC paradigms. This brings data computation and storage capacities to the edge of the mobile networks, and also enables the service orchestration systems that offer mobility and dynamic resource provisioning.

Strong Aspects: Support latency-sensitive IoT applications and services like driverless autonomous vehicles and VR and AR applications. MEC in the entire three-tier computing con-

tinuum accommodates more downlink traffic for IoT services and improves overall network efficiency.

Weak Aspects: The orchestration protocols are hard to define across the networks in MEC, and the computing resources at the MEC nodes are limited as compared to the cloud.

4) *Service Orchestration in Local EC:* Important developments about the latency of the communication data have become low by computing locally at the edge. However, it enhances data privacy and security due to the proximity to the source of data. The orchestration systems developed for resource-constrained edge devices enable the decision-making and processing of data locally at the edge, close to the user.

Strong Aspects: Instant data computation processing is important for real-time IoT applications and services. Data security and privacy are enhanced at the local edge, enabling control and monitoring of data transfer to external servers.

Weak Aspects: As resources are limited at local edge nodes, deploying large-scale IoT applications can be complex. However, managing the network connectivity of a large number of IoT and local edge devices with limited computing capacity, alongside the possible intermittent availability of resources, is also complex.

5) *Service Orchestration in Three-Tier Edge–Cloud Continuum With 6G:* The three-tier computing architecture facilitates a combined computing continuum by integrating the IoT devices (tier 1), MEC with EC/FC nodes (tier 2), and cloud data centers (tier 3), as presented in Fig. 1. The objective of this three-tier edge–cloud continuum architecture is to enhance system performance and the utilization of computing node resources at each tier by facilitating appropriate application and service deployment decisions by the orchestrator. It also makes service provisioning more robust and resilient in unpredictable circumstances such as hardware failure, network outages, power cuts, and security breaches.

Strong Aspects: This involves designing and implementing intelligent service orchestration in the three-tier computing continuum to dynamically handle workload distribution and resource allocation across all three tiers. This ensures that IoT applications and services remain compatible and continue to perform, even in extreme circumstances such as power outages or network disruptions.

Weak Aspects: The major challenge lies in coordinating applications and services across intertier and intratier offloading and communication. Security and privacy threats such as data confidentiality, integrity, trust, and availability also pose a significant challenge when offloading a service to another tier of computing infrastructure.

C. Future Research Directions

Hybrid systems that integrate EC with cloud services can be considered to reduce latency and enhance the QoS performance of time-critical applications. Intelligent resource management by embedding advanced AI and ML enhances the reliability and efficiency of service orchestration in FC. Intelligent service orchestration frameworks supported by blockchain and AI can enable both intratier and intertier service offloading while ensuring consistent application performance across the computing continuum. However, executing

a variety of IoT applications with diverse latency, bandwidth, resource, QoS, and reliability requirements in traditional IoT–cloud, FC, MEC, and EC environments, as well as at the local edge, requires such intelligent orchestration frameworks to ensure efficient service delivery.

The research gap between uniform and standard techniques such as computation offloading, load balancing, task scheduling, and network slicing, in maximizing multitier resource utilization while ensuring reliable QoS is a significant issue. This can only consider IoT services that require high bandwidth and are latency-sensitive. Resource optimization in 6G is a promising area, where the use of computing resources, data storage, and network communication necessitates advanced methods for dynamic resource distribution in edge–cloud continuum architecture using AI and ML techniques. This ensures the stability of the system remains operational while diverse and limited computing resources among multitier computing nodes. Adaptive security systems need to be deployed at EC (tiers 1 and 2) to ensure the security and privacy of the user's application data, where resources are limited. It also ensures a balance between interoperability, resource efficiency of the systems, and privacy protection of the users.

In the context of 6G networks utilizing multitier edge–cloud continuum architectures, the sustained deployment of services is of paramount importance for ensuring the system's performance and reliability. In the event of exceptional situations, such as power outages, network breakages, or security attacks that result in partial interruptions, it is imperative that continuity of operations for services and the safety of data are ensured. A dedicated study is required to investigate resilience against such exceptional circumstances, with the objective of identifying, rectifying, and recuperating from these types of failures in real time.

VI. CONCLUSION

Combining CC with fog and EC is crucial for realizing the full potential of 6G networks. In order to support the 6G computing continuum, this article provides a comparative analysis of the architectural frameworks that currently exist, as well as investigating the recently developed computing paradigms, design concepts, development concepts, and methodologies of both academia and industry. Incorporating intuitions from the literature, we highlighted a multitier edge–cloud continuum architecture as an inclusive framework for orchestrating IoT applications and services, empowering enhanced system performance, maximum resource availability, and computation brought closer to the data sources. This article analyses the reliability and limitations of existing computing frameworks in terms of IoT service orchestration, application requirements, system scalability, latency, resource efficiency, security and privacy, robustness and resilience, and availability at the local edge, EC, FC, MEC, and CC. It discusses recommended methods for deploying 6G networks, emphasizing the importance of consistent definitions and strategies to unify FC, EC, and CC within the 6G ecosystem. This article offers practical insights for future research into the development of 6G networks. It also provides strategic guidance on how to effectively and sustainably meet the needs of the three-tier EC–CC continuum,

establishing a standard for those seeking to utilize 6G network technology to its fullest potential.

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